

Sentinel

Sentinel audits a Sparkplug estate's broker, certificates, identities and topic permissions and grades what it finds, read-only, with evidence and a fix for every finding.

Version 0.1.0 Ignition 8.1.19+ and 8.3 Runs on: any host with broker access

Contents

1. [1. Overview](#)
2. [2. What You Need](#)
3. [3. Install](#)
4. [4. License](#)
5. [5. Configure](#)
6. [6. Operate](#)
7. [7. Verify](#)
8. [8. Troubleshoot](#)
9. [9. Reference](#)
10. [10. Support](#)

1. Overview

Sentinel audits a Sparkplug estate's posture and grades it without ever touching it. It connects to the broker, listens for a bounded window, optionally runs active probes, and produces a scored report where every finding carries the frames or config lines that prove it and a remediation that names the setting. It exposes no publish method at all; it cannot write to the estate.

Sentinel is vendor-neutral: it audits any Sparkplug estate, not only one running SparkBridge.

2. What You Need

- A SparkBridge Host gateway for the estate being audited.
- The signed Sentinel release from the SparkBridge bundle (built as `sparkaudit-<version>-all.jar`).
- Java 17 if building from source.
- Broker credentials for the auditor identity; optionally the broker's `credentials.xml` and a posture properties file for config-derived findings.

3. Install

Sentinel carries its own license. Install the signed release through Config > Modules > Install or Upgrade a Module; no build step is required for the packaged release.

To build from source: requires Java 17 and `io.sparkbridge:common` published to mavenLocal.

```
JAVA_HOME=~/.local/opt/jdk-17.0.19+10/Contents/Home ./gradlew fatJar
java -jar build/libs/sparkaudit-0.1.0-all.jar --help
```

4. License

Soft-licensed; Sentinel runs in full whether or not a key is present. License file lookup order: `--license <file>`, the `SPARKBRIDGE_LICENSE` environment variable, then `sparkbridge-license.key` in the working directory. Run `sparkaudit license` for full status.

5. Configure

Sentinel runs as a passive audit by default:

```
java -jar sparkaudit-0.1.0-all.jar \
  --broker tcp://broker.example:1883 --username auditor \
  --password-file secrets.properties audit.password \
  --estate "Plant 4" --window 300 --expect-groups Plant4,Utilities \
  --out-html report.html --out-json report.json
```

--credentials-xml, --posture Add config-derived findings the wire alone cannot show.

--allow-active-probes Enables active probes. Off by default.

--tls-listener Checks a TLS listener.

--profile free / --profile full `free` withholds full evidence for a pre-sales report; `full` (default) includes complete evidence and remediation.

--capture-out capture.jsonl Writes the raw window to disk.

--replay capture.jsonl Re-evaluates rules offline against a captured window, no broker involved.

The four axes Sentinel scores: connection posture, and (per the module's own documentation) further axes covering broker configuration, identity and permission hygiene, and topic/namespace isolation. Active probes only run with `--allow-active-probes`; with probes off, the connection-posture axis reports that it was not exercised.

6. Operate

Sentinel is a point-in-time audit tool, not a running service. Run it on demand or on a schedule against an estate's broker. It never publishes to any topic and exposes no publish method. Use `--capture-out` to keep a raw record of any audit window so a finding stays checkable later, or by someone who was not in the room, or so a customer can send an estate SparkBridge is not allowed to connect to.

7. Verify

Run the audit and check the exit code: 0 clean, 1 findings below critical, 2 at least one critical finding, 3 the audit could not run. Use `--capture-out capture.jsonl` and later `--replay capture.jsonl` to confirm the same findings reproduce offline against the same window.

8. Troubleshoot

Symptom	Cause	Fix
Exit code 3	The audit could not run (bad broker URI, bad credentials, or the window elapsed with no connection)	Confirm <code>--broker</code> , <code>--username</code> and the password file path are correct

Symptom	Cause	Fix
Connection-posture axis reports "not exercised"	Active probes are off by default	Add <code>--allow-active-probes</code> if authorized to probe the estate
Missing config-derived findings	<code>--credentials-xml</code> or <code>--posture</code> not supplied	Pass the broker's real <code>credentials.xml</code> and a posture properties file
build/libs jar missing when building from source	<code>io.sparkbridge:common</code> not published to <code>mavenLocal</code>	Run <code>./gradlew :common:publishToMavenLocal</code> in the SparkBridge repo first

9. Reference

- Exit codes: 0 clean, 1 sub-critical findings, 2 critical finding, 3 audit could not run.
- License file: `sparkbridge-license.key` (working directory, `--license`, or `SPARKBRIDGE_LICENSE`).
- Build artifact: `sparkaudit-0.1.0-all.jar`.

10. Support

[Sentinel in the docs](#) · [Contact](#) · [GitHub releases](#)